Persistent Homology of Music Network with Three Different Distances

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Project page

The big idea

Distance is a **tunable resolution knob** on musical structure — sweeping $d_1 \to d_3 \to d_2$ distills a piece down to its **core cyclic skeleton**.

- **Three path distances**: d_1 (min edges), d_2 (min weight — the only metric), d_3 (hybrid).
- **Proven ordering** $d_2 \le d_3 \le d_1$ lifts to a **nontrivial barcode injection** $\mathrm{bcd}_1(d_2) \hookrightarrow \mathrm{bcd}_1(d_3) \hookrightarrow \mathrm{bcd}_1(d_1)$ (Heo–Choi–Jung 2025).
- **Validated** on real Korean court music — an exact match to the paper's Table 1.

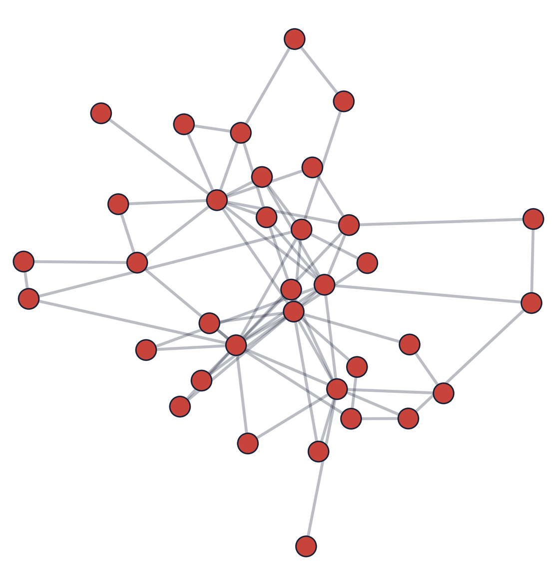
Cycle skeleton extraction

d_1: 7 cycles d_3: 4 cycles d_2: 3 cycles

On *Sanghyeondodeuri* (geomungo): as $d_1 \to d_3 \to d_2$, the H_1 barcode thins $|\mathrm{bcd}_1| = \mathbf{7} \to \mathbf{4} \to \mathbf{3}$ — only the robust core loops survive.

From music to a graph

- Each **node** = (pitch, duration);
- an **edge** joins consecutive notes;
- the **weight** is 1/(co-occurrence count) — frequent neighbors are closer.



The music network of *Sanghyeondodeuri* (geomungo).

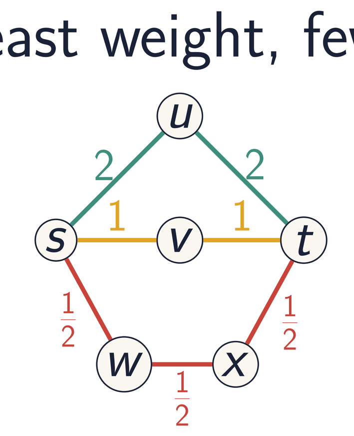
Three path distances

hops $\ell(P)$ of Path P, $h^* = \min_P \ell(P)$;
weight $\omega(P) = \sum_{e \in P} W(e)$.

$d_1 = \omega(P)$, $\ell(P) = h^*$ (Dijkstra alg.) — a fewest-edge path.

$d_2 = \min_P \omega(P)$ — least weight; *the only metric*.

$d_3 = \min\{\omega(P) : \ell(P) = h^*\}$ — hybrid (least weight, fewest edges).



$s \to t$: $d_1{=}4 > d_3{=}2 > d_2{=}1.5$. Both $s\,u\,t$ and $s\,v\,t$ are fewest-edge; d_1 follows the one picked by **vertex order** (u before v), d_3 the lightest, d_2 the lightest.

Only d_2 is a metric

d_2 satisfies the triangle inequality; d_1 and d_3 can violate it (a heavy direct edge). Yet all three give meaningful musical structure.

A universal ordering

For *every* pair of nodes,

$$d_2(v,w) \le d_3(v,w) \le d_1(v,w).$$

This universal ordering induces an injection in one-dimensional persistent homology

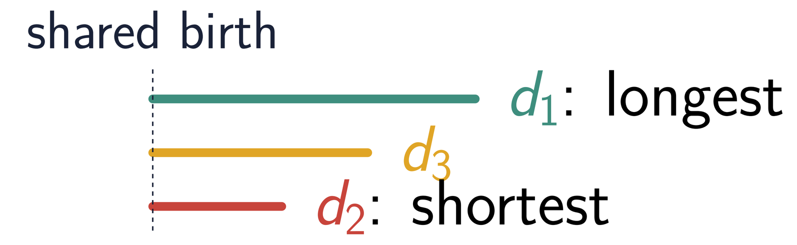
Barcode injection

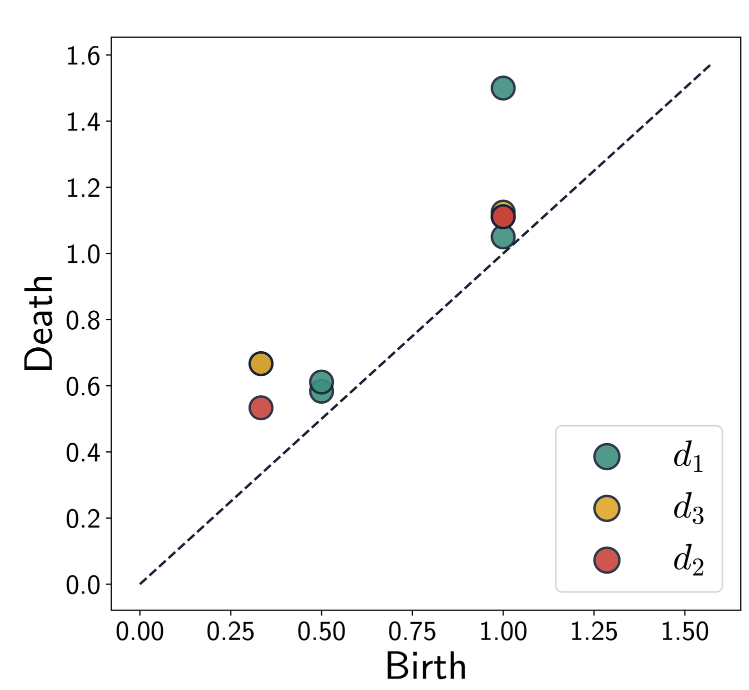
The birth edges nest, $B_2 \subseteq B_3 \subseteq B_1$, give injections

$$\mathrm{bcd}_1(d_2) \hookrightarrow \mathrm{bcd}_1(d_3) \hookrightarrow \mathrm{bcd}_1(d_1).$$

by **nontrivial theorem** for path-representable distances (**Heo–Choi–Jung 2025**).

Each cycle keeps its **birth**; its **lifespan grows** $d_2 \le d_3 \le d_1$, and the number of cycles can only increase.



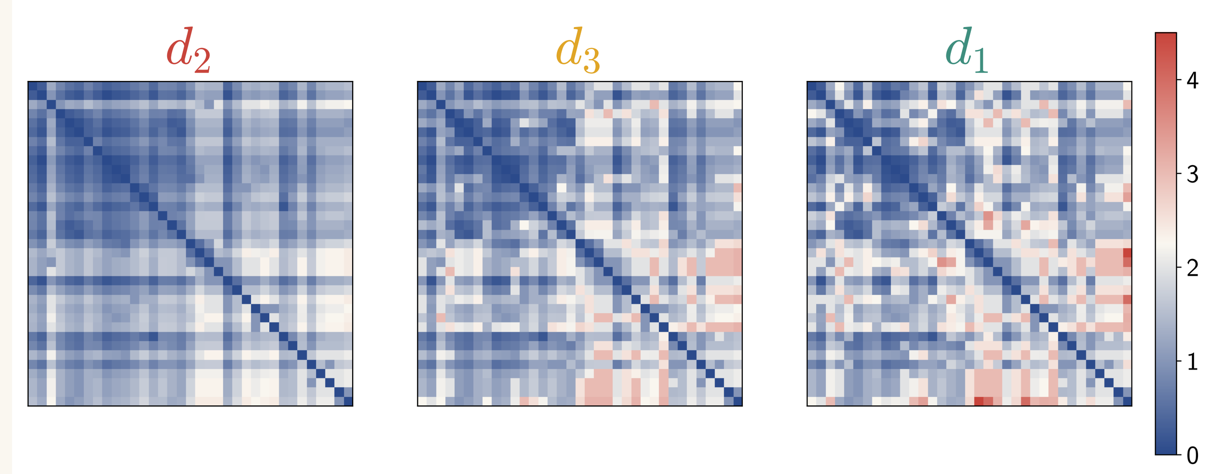


On real music: d_1 points lie above d_2 — longer lifespans, shared births.

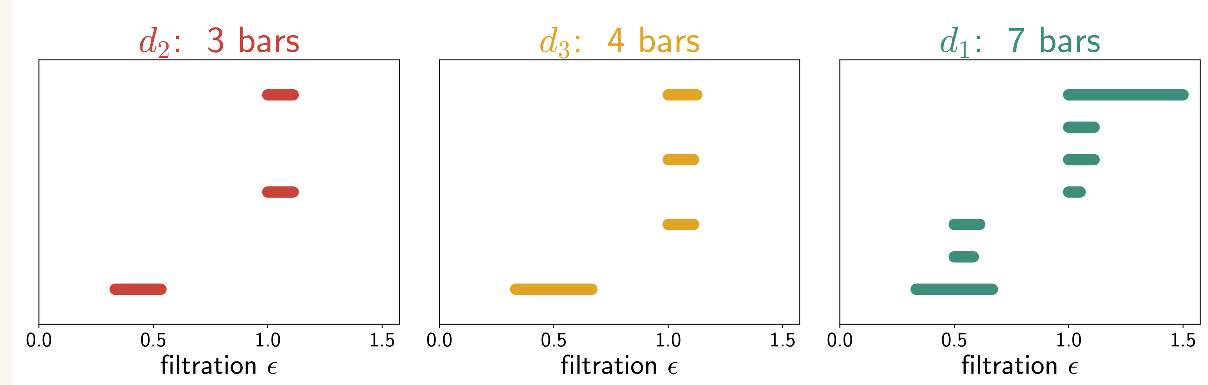
Why it matters

The same piece yields *different but nested* topological readings. d_2 exposes the robust core; d_1 adds finer, peripheral loops. The changes are **systematic, not random**.

Evidence on real music



Distance matrices: d_2 (cool/smooth) $\to$ d_1 (hot) the ordering is visible.



H_1 barcodes: $|\mathrm{bcd}_1(d_1)|, |\mathrm{bcd}_1(d_3)|, |\mathrm{bcd}_1(d_2)| = \mathbf{7}, \mathbf{4}, \mathbf{3}$; lifespans grow, births shared.

Distilling the core

Ranking d_1 cycles by how far down they survive ($\to d_3 \to d_2$) gives a principled **importance hierarchy of musical loops** — a multi-resolution descriptor of a piece.

Takeaways & future work

- Distance choice = a tunable lens; results are nested, not arbitrary.
- Toward **AI music composition** with multiple stylistic perspectives from one source.

Interactive explorer & audio demos on the project page.

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Key refs: **Heo–Choi–Jung (2025) arXiv:2501.03553** (barcode injection); Tran–Park–Jung (2023) *J. Math. Music*; Sethares–Budney (2014).
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